AI CLUSTERS

System Efficiency for Real-World AI Inference: Arm and FuriosaAI for Scalable Open Infrastructure

Satadal Bhattacharjee

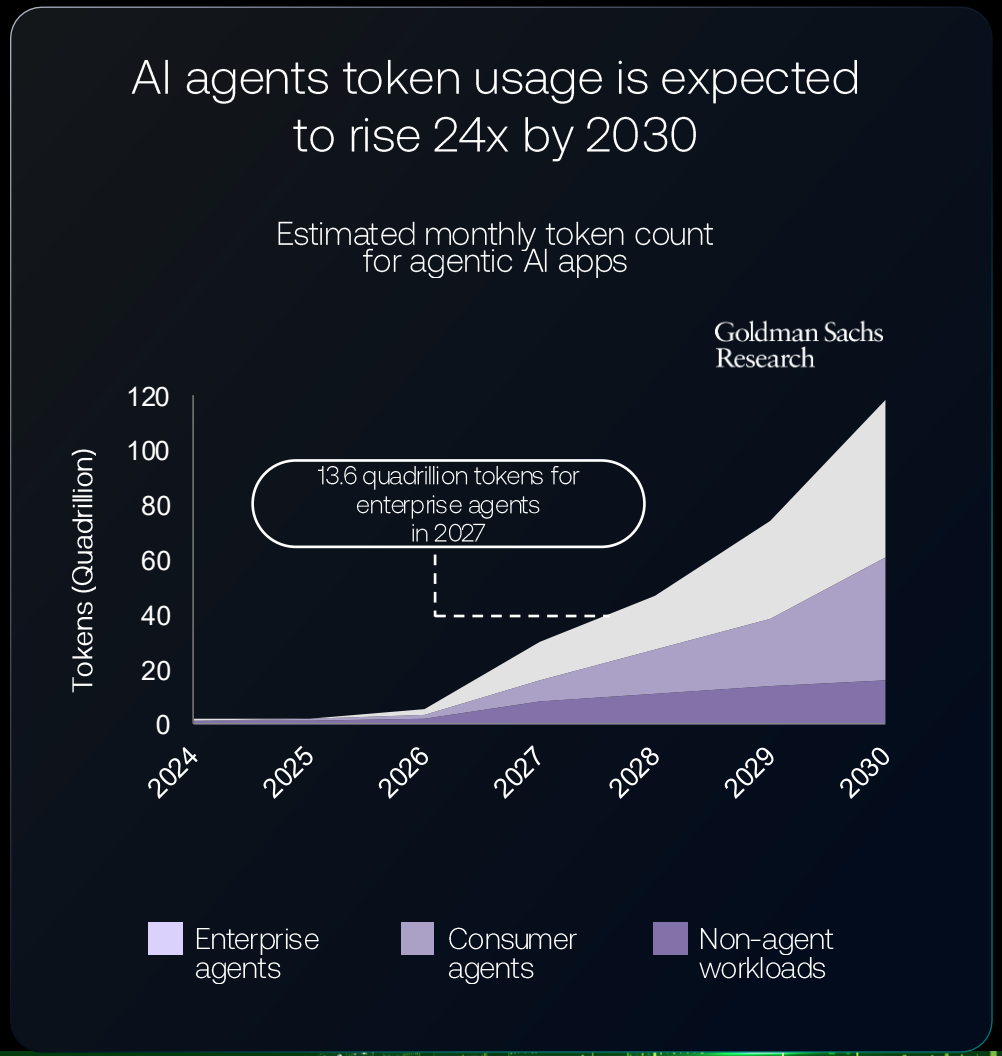
Global Head of Cloud and AI Infra Silicon, Arm

Jihoon Yoon

Product Manager, FuriosaAI

OCP PLATINUM Member

Agents drive CPU demand



CPU

ACCELERATOR

Task distribution

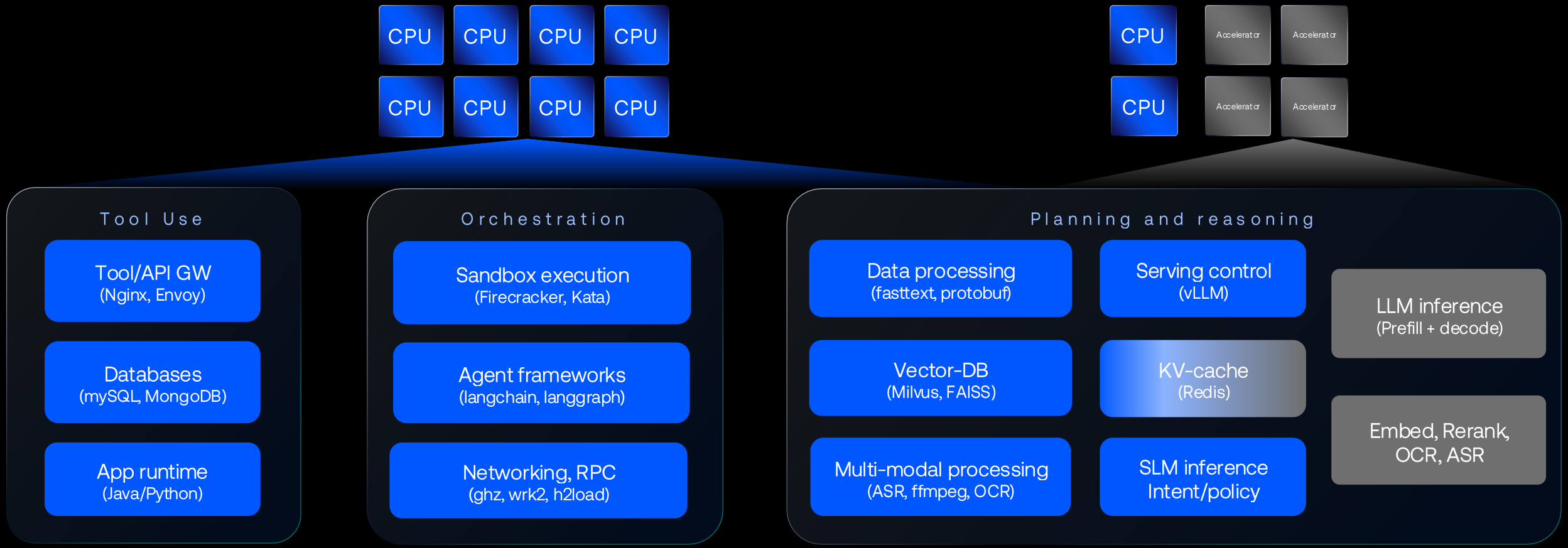
Enterprise automation
Dominated by concurrent interactions
50% | 50%

Software development
Dominated by dependency chains
60% | 40%

Financial analysis
Dominated by parallel decomposition
40% | 60%

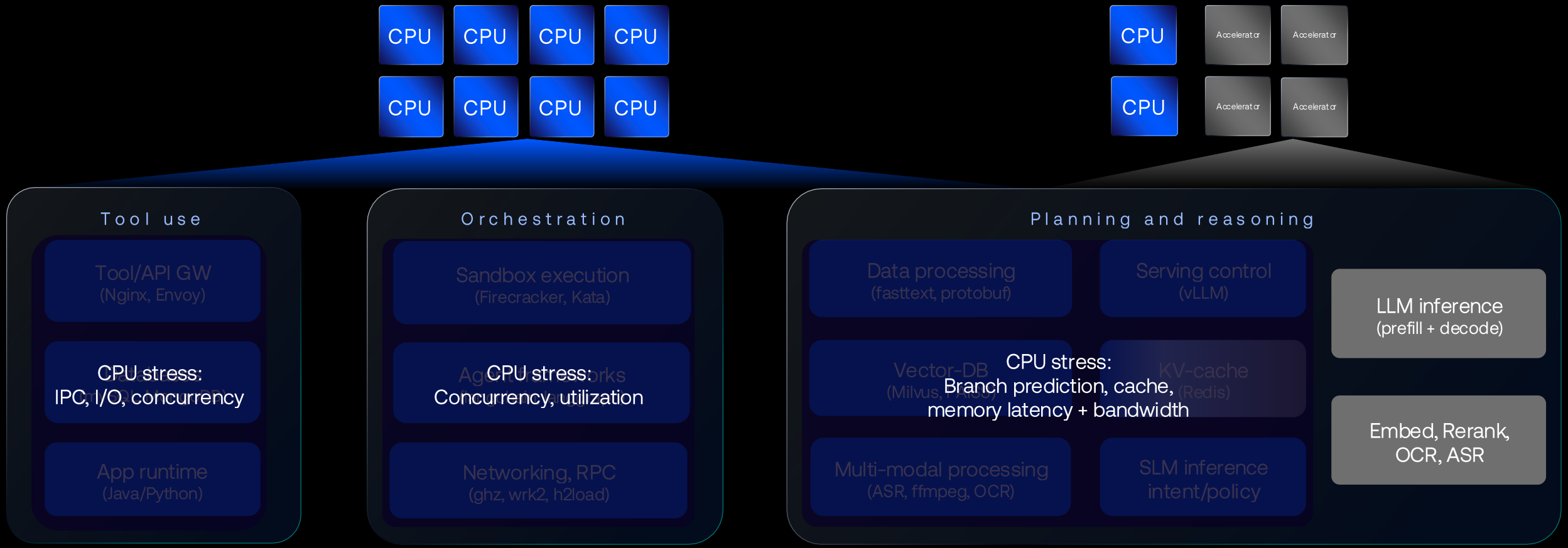
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Agentic Workload Performance



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Agentic Workload Performance

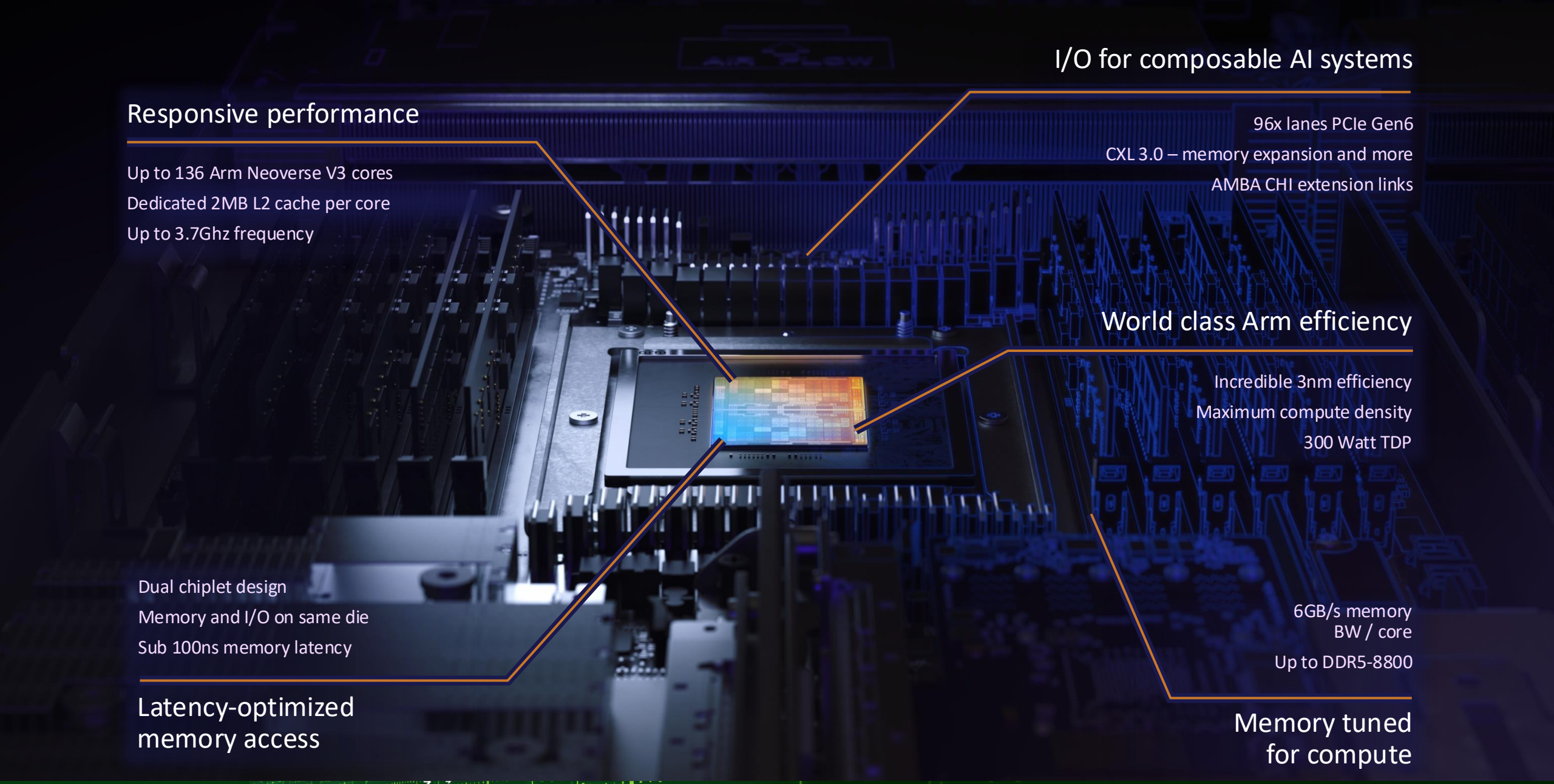


Key agentic performance indicators:

- Worst case (p99) performance at each step (tools, orchestration, planning/reasoning)
- Start-up/switching latency and steady-state throughput

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arm AGI CPU
World's most efficient agentic CPU
Deployment Partners
cerebras
CLOUDFLARE
f5
Meta
OpenAI
ORACLE
SAP
SK telecom
verda
火山引擎 VolcanoEngine
OCP GLOBAL SUMMIT 2026
Scaling Innovation for the AI Era.



Responsive performance

Up to 136 Arm Neoverse V3 cores
Dedicated 2MB L2 cache per core
Up to 3.7Ghz frequency

I/O for composable AI systems

96x lanes PCIe Gen6
CXL 3.0 – memory expansion and more
AMBA CHI extension links

World class Arm efficiency

Incredible 3nm efficiency
Maximum compute density
300 Watt TDP

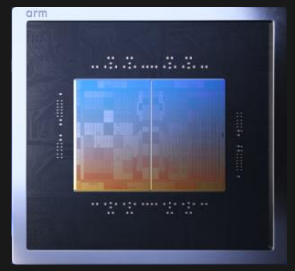
Latency-optimized memory access

Dual chiplet design
Memory and I/O on same die
Sub 100ns memory latency

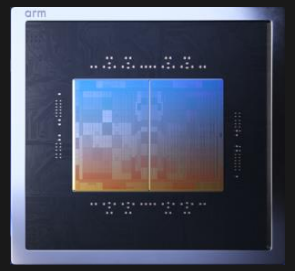
Memory tuned for compute

6GB/s memory BW / core
Up to DDR5-8800

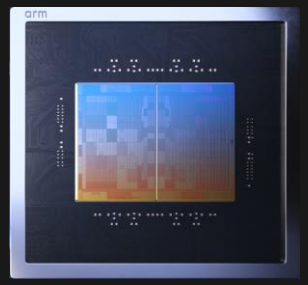
Arm AGI CPU: Product Family



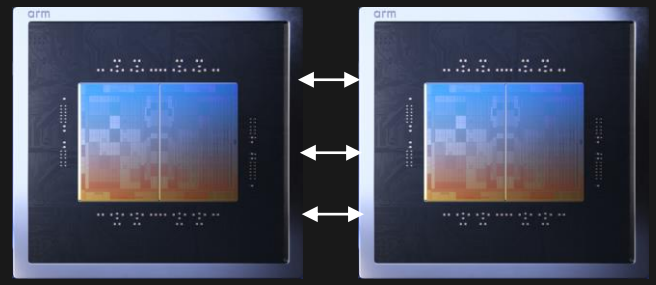
64 Cores
3.7 GHz Boost
96x PCIe Gen6



128 Cores
3.2 GHz Nominal
96x PCIe Gen6



136 Cores
3.5 GHz Boost
96x PCIe Gen6

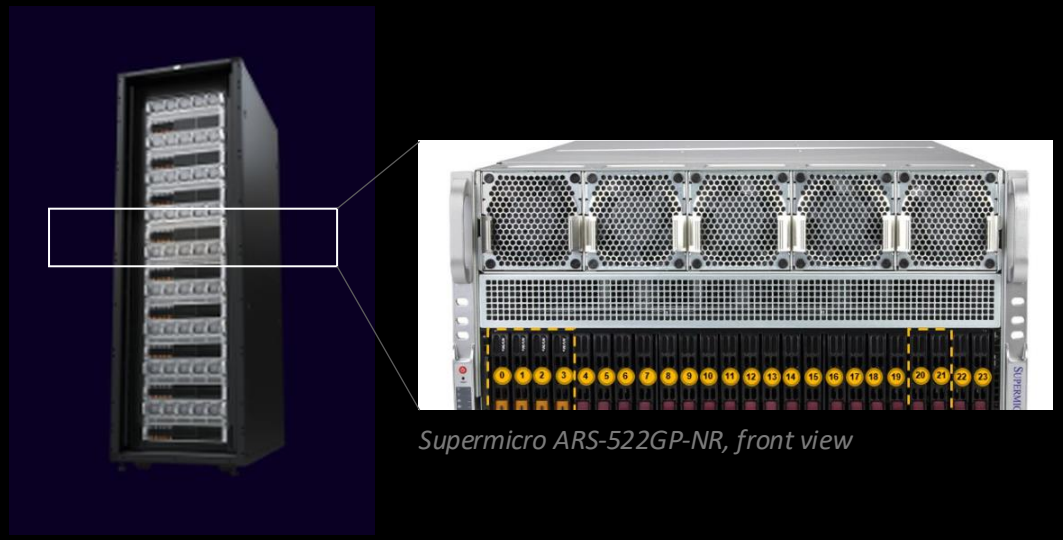


272 Cores (Dual SoC)
24x DDR5-8800
128x PCIe Gen 6

Leading Perf/W for agentic and cloud-native workloads

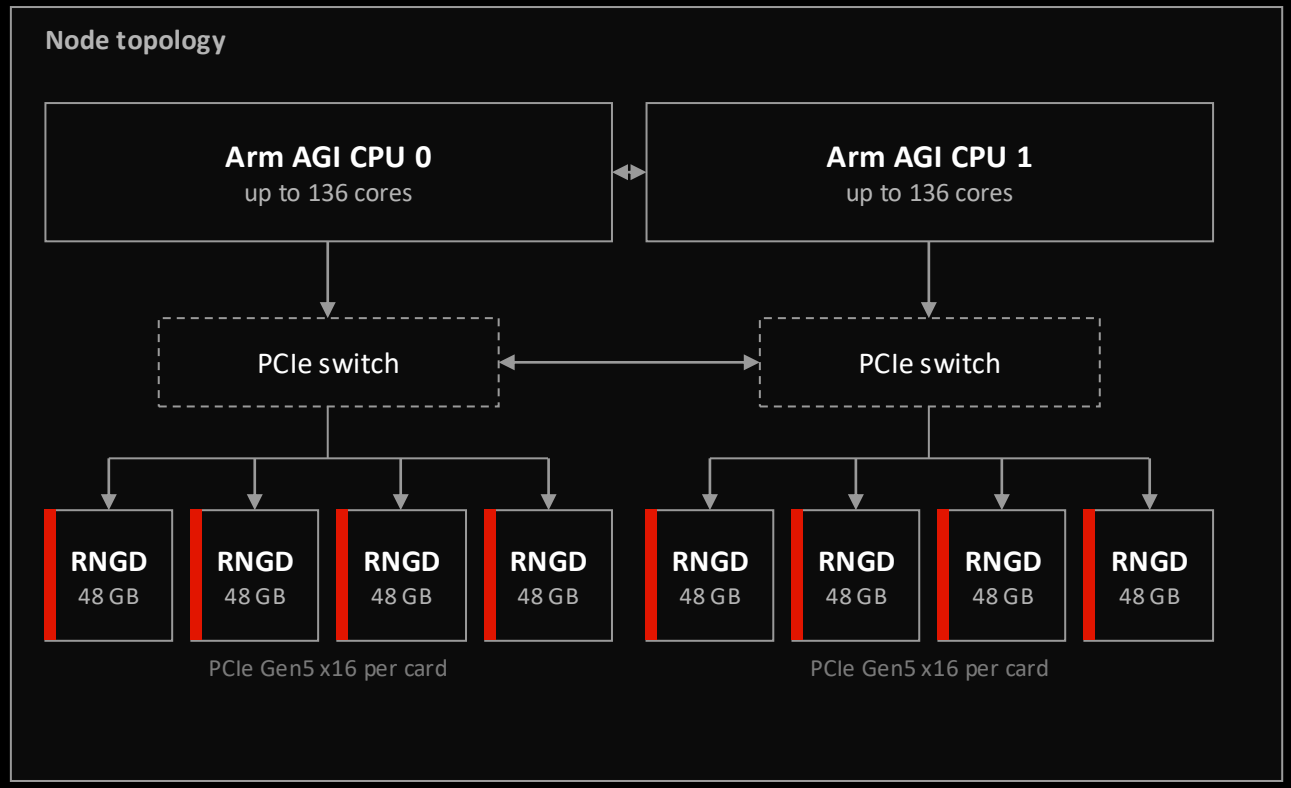
Tight coupling between CPU and AI to boost platform performance

Target system: Supermicro ARS-522GP-NR



Supermicro ARS-522GP-NR, front view

- 2 × Arm AGI CPU, up to 136 Neoverse V3 cores each, 300 W
- 24 × DDR5 RDIMM, 12 channels per socket, up to 6 TB
- 8 × PCIe Gen5 x16 double-width slots → 8 × RNGD
- OCP NIC 3.0 slot · 6 × 2,700 W PSU · 19-inch, air-cooled



FuriosaAI builds powerful and energy-efficient accelerators for cloud and sovereign AI

AI inference with
energy efficiency at its core



Founded in 2017

RNGD: purpose-built for energy-efficient inference

Two generations of inference silicon; RNGD in volume production and deployed from single cards to full racks.

RNGD card



512 TFLOPS FP8

48 GB HBM3 · 1.5 TB/s

180 W · passive · PCIe Gen5

NXT RNGD Server



8 cards · 4 PFLOPS FP8

384 GB HBM3

3 kW · 4U · air-cooled

RNGD rack



8 servers · 32 PFLOPS FP8

3.07 TB HBM3 · 96 TB/s

< 30 kW · air-cooled

Why we are validating RNGD on Arm AGI CPU

Customers want options for a fast-changing mix of workloads, and at inference scale energy efficiency decides capacity.

01

Use cases multiply and evolve fast

Agents, retrieval, multimodal and voice pipelines each stress different resources. Workload-specific needs call for more than one kind of compute.

02

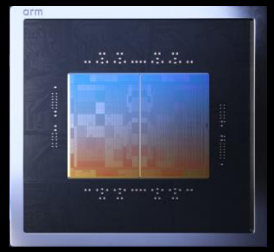
Customers asking for heterogeneous infrastructure

A real choice of CPU, accelerator and form factor on open standards (PCIe, OCP), not one vendor's closed stack.

03

Inference runs at scale, so efficiency sets capacity

Rack power is fixed. System-level tokens per watt, host CPU plus accelerator, decides how much inference a rack can serve.



Arm AGI CPU

×



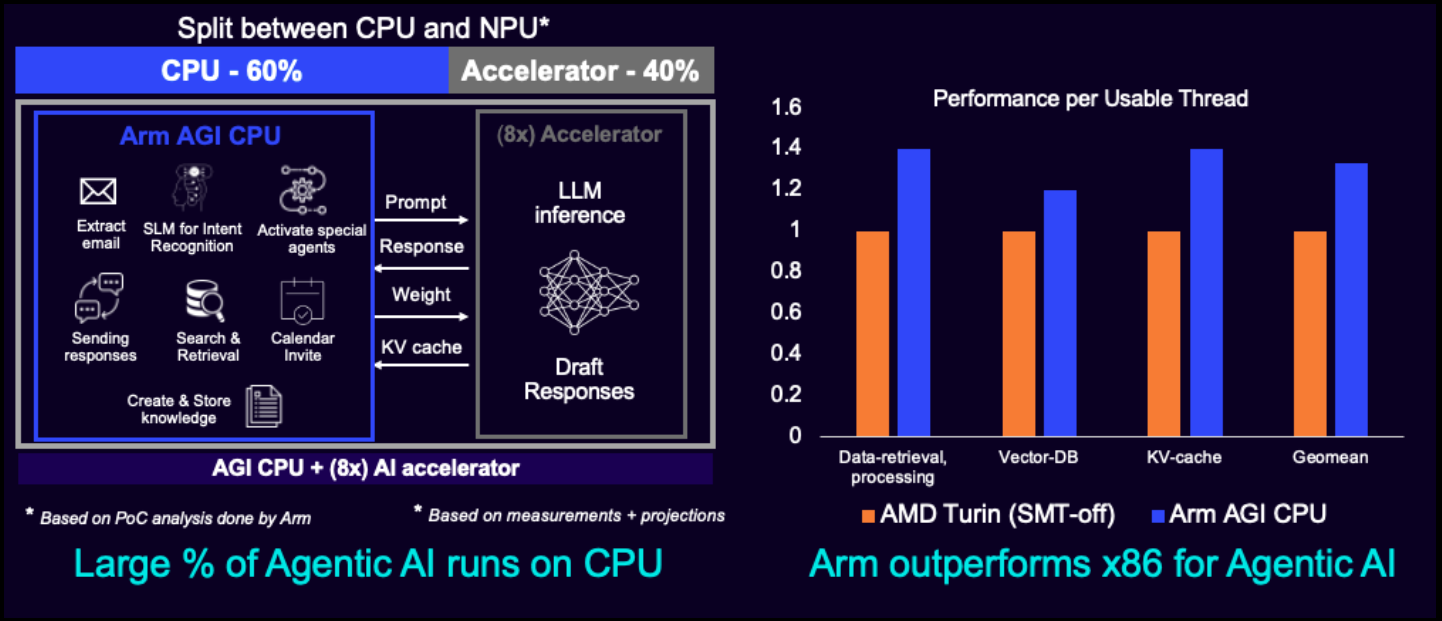
RNGD

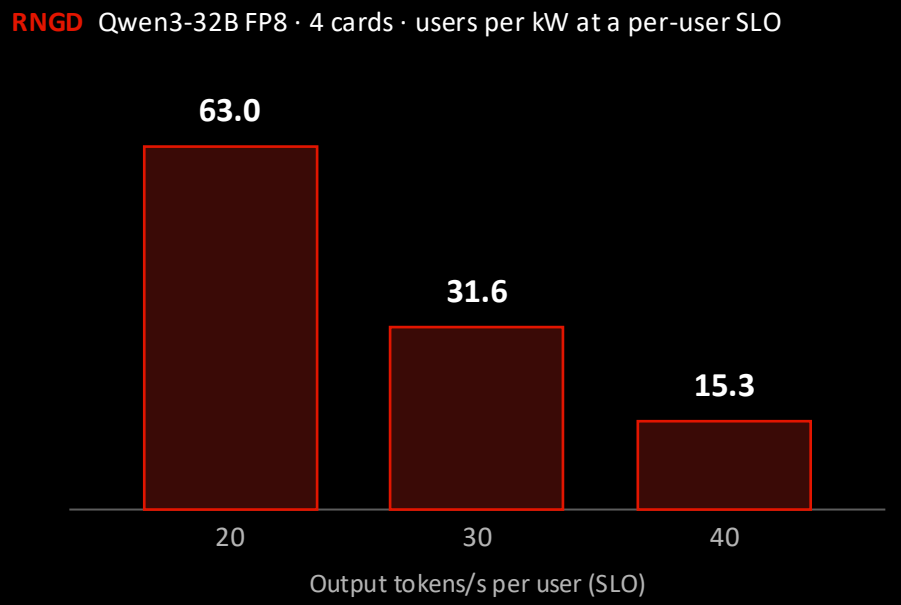


Supermicro ARS-522GP-NR

Dual Arm AGI CPU + 8× RNGD in one 5U, air-cooled node

Agentic inference pairs CPU workflow stages with SLO-dependent LLM serving





1K/1K tokens, vLLM-compatible serving, x86 host; no Arm-host result yet.

CPU side (Arm): about 60% of the agent's steps run on the CPU; on those steps Arm AGI CPU delivers 1.2–1.4× per usable thread vs. the x86 baseline.

LLM side (RNGD): the latency target sets capacity: 63 users per kW at 20 tok/s, 15 at 40. At 30 tok/s per user, 4 cards serve ~1,950 tok/s (TTFT 0.35 s).

Rack-scale systems need software readiness



Serve across nodes

Disaggregated prefill/decode and request routing across the rack (llm-d)

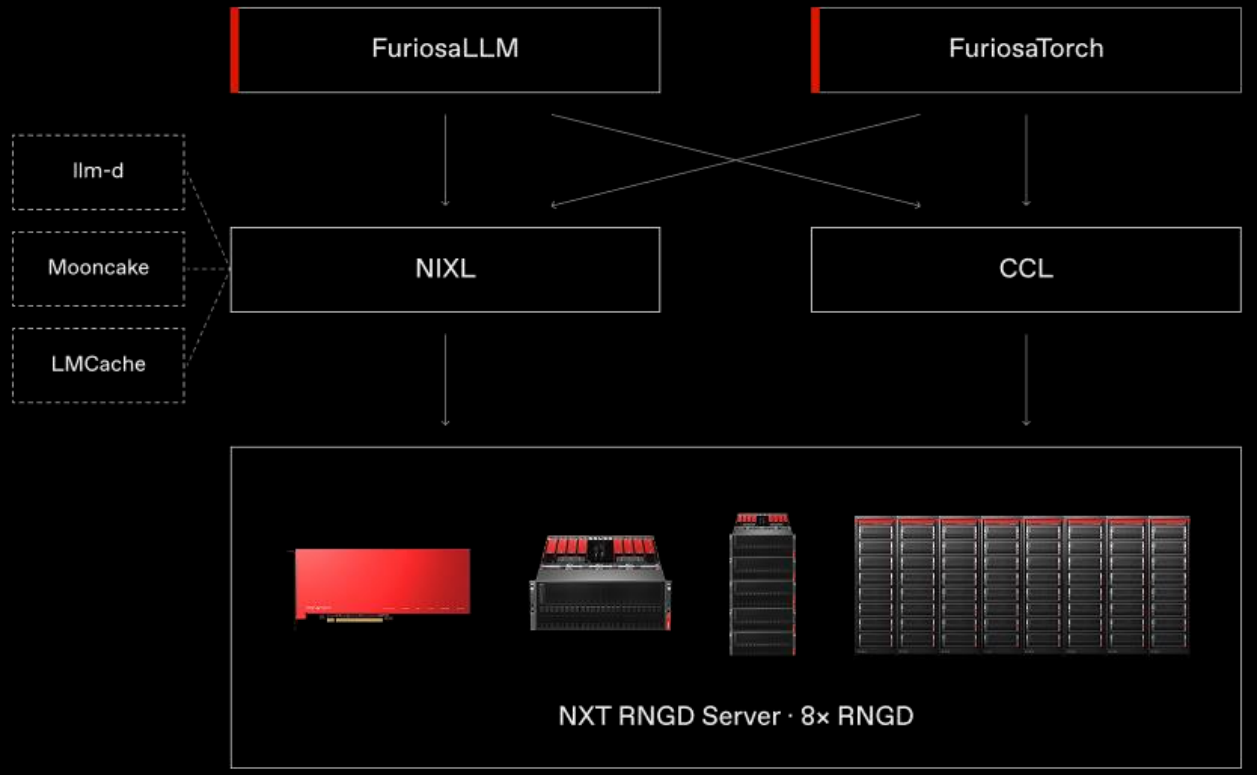
Move and tier KV cache

Across HBM, host DRAM and nodes via NIXL, with Mooncake and LMCache backends

Collectives across cards and nodes

Tensor, pipeline and data parallelism over Collective Communication Library (CCL)

Leveraging opensource serving stack



30 kW ORV3 rack: 2,176 cores, 32 PFLOPS, 3 TB HBM, air-cooled



2,176
Neoverse V3 cores
16 x 136

32 PFLOPS
FP8 compute
64 × 512 TFLOPS

3.07 TB
HBM3 capacity
64 × 48 GB

< 30 kW
rack power budget
8 nodes at ~2.5–3 kW + switching

The new joint rack will able to bring compute and memory-dense inference compute for new and existing infrastructure.

Takeaways

01 **Agentic inference is a pipeline, not a model call.**
Tool use, orchestration and retrieval on the CPU set end-to-end latency and cost as much as the accelerator stage does.

02 **Match each stage to its unit.**
CPU, AI Accelerator card, node or rack, match to their respective boundary, SLO and duration for optimal efficiency at the rack level.

03 **Heterogeneous, open, air-cooled nodes scale from today's racks up.**
An efficient CPU and an efficient accelerator in a standard PCIe node fit existing 30 kW racks and become the unit for denser designs.

Get involved

Arm AGI CPU
arm.com/products/cloud-datacenter/arm-agi-cpu

Arm AGI CPU Agentic Benchmarks

FuriosaAI RNGD
developer.furiosa.ai · contact@furiosa.ai

System efficiency comes from matching each workload stage to the right compute unit and co-designing the path from CPU and accelerator through node, rack, software and operations.